Unit 5: Impact of Global Warming on Pakistan

Complete exercise solutions — all textbook questions answered (full book, 14 units)

Reading and Critical Thinking — Exercise A (Answers)

Q1. What makes Pakistan particularly vulnerable to the impacts of global warming?

• Pakistan's vulnerability is largely due to its diverse geography — high mountain ranges, vast plains, and an extensive coastline — which makes it prone to glacial melt, extreme heat, and sea-level rise.

• Heavy dependence on agriculture, rapid urbanisation with inadequate infrastructure, limited financial and technical resources, and a population exceeding 220 million further intensify the effects.

Q2. How does rising temperature affect urban areas like Karachi?

• Rising temperatures cause more frequent and intense heatwaves in urban areas like Karachi, where rapid growth has outpaced infrastructure development. Urban heat islands make densely built-up areas retain more heat.

• Such extreme heat poses severe health risks, as evidenced by the 2015 Karachi heatwave, which resulted in over 1,200 deaths. The primary cause is the enhanced greenhouse effect.

Q3. What is the connection between glacial melting and water scarcity in Pakistan?

• Glaciers in the Himalayas and Karakoram feed the Indus River, on which agriculture and drinking water depend. Accelerated melting initially increases water flow (floods), but over time it results in reduced water availability.

• Over 30% of the ice has already melted, leading to reduced river flows, which threatens agriculture and drinking water supplies.

Q4. Why are extreme weather events becoming more frequent and severe in Pakistan?

• Global warming has disrupted weather patterns, leading to more intense rainfall and prolonged dry periods. Pakistan's geographical diversity, rapid urbanisation, and population growth make it susceptible to floods, droughts, and cyclones.

• The 2010 floods affected over 20 million people and caused $10 billion in losses, while Tharparkar faces frequent droughts. The increased frequency is largely due to changes in atmospheric conditions caused by global warming.

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Q5. How does global warming affect agriculture and food security in Pakistan?

• Changes in temperature, precipitation, and extreme weather have made agriculture more unpredictable and less productive. Wheat yields are projected to decline 6–8% by 2050.

• The 2020 locust attack, driven by unusual weather patterns, caused crop losses worth $2.2 billion. Pakistan is promoting Climate-Smart Agriculture — drought-resistant crops, optimised water use, and better soil management.

Q6. Why is it important for Pakistan to implement the Heatwave Management Plan under SDG 13?

• The Heatwave Management Plan includes early warning systems and public awareness campaigns. It protects lives by alerting people before dangerous heatwaves, reducing deaths like the 1,200 in Karachi in 2015.

• It is part of Pakistan's commitment to SDG 13 (Climate Action), which requires urgent action to combat climate change and its impacts.

Q7. What role does Pakistan's reliance on agriculture play in its vulnerability to climate change?

• Agriculture is highly sensitive to changes in temperature and precipitation. Since Pakistan depends heavily on this sector for food security and employment, any climate shift directly threatens crops, livestock, and livelihoods.

• A bad season can mean food shortages, higher prices, and economic hardship for millions.

Q8. Can limited financial and technical resources hinder Pakistan's ability to adapt to climate change?

• Yes. Adaptation requires investment in infrastructure, early warning systems, water management, and new technologies. Limited resources slow down projects like the GLOF early warning system, flood defences, and climate-smart agriculture.

• Without adequate resources, vulnerable communities remain exposed to disasters.

Exercise B — Choose the Correct Option (Answer Key)

No.	Question	Answer
1	Which of the following is a major reason for Pakistan's vulnerability to climate change?	b. Geographic diversity
2	What is the main effect of glacial melting in Pakistan as mentioned in the text?	c. Reduced water availability
3	Which city in Pakistan is particularly affected by rising temperatures and heatwaves?	b. Karachi
4	Which step has Pakistan taken to address the risks of glacial melting?	c. Glacial Lake Outburst Flood (GLOF) Project
5	Which areas are being threatened by the rising sea levels in Pakistan?	c. Coastal regions

Exercise C — Mind Map (Model)

• **Central idea:** "Impact of Global Warming on Pakistan."

• **Rising Temperatures:** +0.6°C (2001–2020); Jacobabad 51°C; Karachi heatwave 2015 (1,200 deaths); Heatwave Management Plan.

• **Glacier Melting:** over 30% of ice melted; Indus River flow reduced; GLOF project.

• **Extreme Weather Events:** 2010 floods (20 million affected, $10 billion losses); Tharparkar droughts; National Disaster Risk Management Framework.

• **Agricultural Impact:** wheat yields to decline 6–8% by 2050; 2020 locust attack ($2.2 billion); Climate-Smart Agriculture.

• **Coastal Erosion:** sea level +0.6 m by 2100; 5 million displaced; Mangrove Rehabilitation Project.

Exercise D — Summary (Model, 150–200 words)

Global warming, driven by human activities, is having multifaceted effects on Pakistan, a country highly vulnerable because of its diverse geography, dependence on agriculture, rapid urbanisation, limited resources, and large population. Rising temperatures have increased heatwaves, as shown by Jacobabad's 51°C record and the 2015 Karachi heatwave that killed over 1,200 people. Melting glaciers in the Himalayas and Karakoram have reduced the flow of the Indus River, threatening agriculture and drinking water. Extreme weather events such as the 2010 floods and recurring droughts in Tharparkar have caused huge losses. Agriculture faces declining wheat yields and locust attacks, while sea-level rise threatens the coasts of Sindh and Balochistan. Biodiversity is also declining through habitat loss. Pakistan is responding under SDG 13 through the Heatwave Management Plan, the GLOF project, climate-smart agriculture, the Mangrove Rehabilitation Project, and the Protected Areas Initiative.

Exercise E — Table (Model)

Key Aspect	Description of Impact	SDG 13 Initiative
Rising temperatures	Heatwaves; Jacobabad 51°C; Karachi 2015 deaths	Heatwave Management Plan
Glacier melting	Over 30% of ice melted; Indus flow reduced	GLOF Project
Extreme weather	2010 floods; Tharparkar droughts	National Disaster Risk Management Framework
Agriculture	Wheat yields to decline 6–8% by 2050	Climate-Smart Agriculture
Coastal erosion	Sea level +0.6 m by 2100; 5 million displaced	Mangrove Rehabilitation Project
Biodiversity	43,000 hectares of forest lost annually	Protected Areas Initiative

Exercise F — Data-Based Questions (Answers)

• **Table 1 (Temperature):** a. The temperature increase is rising steadily: +0.2°C (1901–1950), +0.3°C (1951–2000), +0.6°C (2001–2020). b. This trend means more frequent and intense heatwaves in cities like Karachi, with severe health risks.

• **Table 2 (Glaciers):** a. As the melting percentage rises (10% → 20% → 30%), the impact on Indus River flow grows from slight to noticeable to significant reduction. b. Reduced water availability threatens irrigation, crop production, and drinking water supplies.

• **Chart (Extreme weather):** a. The rising trend suggests Pakistan's climate is becoming more unstable and disaster-prone. b. More extreme events damage infrastructure, displace people, and cause heavy economic losses.

• **Table 4 (Sea level):** a. Sea levels are projected to rise steadily (0.2 m by 2020, 0.3 m by 2050, 0.6 m by 2100), threatening coastal areas. b. Up to 5 million people may be

displaced by 2100, with implications for housing, livelihoods, and infrastructure in Sindh and Balochistan.

Vocabulary Exercises (Answers)

• **Context clues:** exacerbated — made worse ("vulnerability is exacerbated by geographic diversity"). fluctuations — irregular rising and falling ("sensitive to fluctuations in temperature").

• **Word forms of "climate":** climate (noun), climatic (adjective), climatically (adverb).

• **Sentences:** The climate of Pakistan is changing rapidly. Climatic changes affect rainfall patterns. The region is climatically vulnerable to drought.

Grammar — Idioms and Proverbs (Answers)

• **Identify:** 1. "Better late than never" — **Proverb**: it is better to do something late than never. 2. "lends a hand" — **Idiom**: helps.

• **Complete:** 1. a) bite the bullet. 2. b) put all his eggs in one basket. 3. c) The early bird catches the worm. 4. b) Rome wasn't built in a day. 5. c) let sleeping dogs lie.

• **Sentences:** 1. The exam was a piece of cake for her. 2. He told a joke to break the ice. 3. Actions speak louder than words, so I will show you. 4. He let the cat out of the bag about the surprise. 5. I can't come today; I'm under the weather. 6. Too many cooks spoil the broth. 7. The car cost an arm and a leg. 8. The ball is in your court now. 9. She burned the midnight oil before the exam. 10. You can't judge a book by its cover.

Grammar — Conjunctions (Answers)

• **Choose:** 1. b) but. 2. a) unless. 3. c) Both / and. 4. b) however. 5. a) While.

• **Identify:** 1. "after" — subordinating. 2. "Not only... but also" — correlative. 3. "therefore" — conjunctive adverb. 4. "yet" — coordinating.

• **Join:** 1. He is not only intelligent but also hardworking. 2. Although the bus was late, we still arrived on time. 3. You must submit the form unless you won't be allowed to take the test. (Correct form: "You must submit the form, or you won't be allowed to take the test" — using "unless": "Unless you submit the form, you won't be allowed to take the test.")

Writing Task — Model Answer

Informative Essay (250–300 words): "The Importance of Climate Action in Addressing Global Warming"

• **Introduction:** Global warming is the defining challenge of our time. Pakistan, with its diverse geography and dependence on agriculture, is among the most vulnerable countries. Climate action under SDG 13 is therefore not a choice but a necessity.

• **Body:** First, rising temperatures and heatwaves threaten human life, as shown by the 2015 Karachi heatwave. Second, melting glaciers reduce the Indus River's flow, endangering agriculture and drinking water. Third, extreme weather events cause massive economic losses, as in the 2010 floods. Pakistan is responding through the Heatwave Management Plan, the GLOF project, climate-smart agriculture, and mangrove rehabilitation. Critics may argue that climate action is expensive, but the cost of inaction

— lost lives, crops, and homes — is far greater.

• **Conclusion:** Climate action is urgent and essential. Every sector — government, business, and ordinary citizens — must act now to protect Pakistan's people and environment for future generations.

Note: Textbook errors flagged: (1) Table 1 cites "Asian Development Bank, 2027" — a future date for 1901–2020 data. (2) Table 5 (Forest Cover Loss) is incomplete in print. (3) The extreme-weather chart is dated "(2009, 2020)" in one place and "(2000–2020)" in another.

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